Department of Computer Applications
23MX21 – Software Project Estimation
Learning Outcomes based questions

1. Apply COCOMO technique to compute all the estimates required for an embedded project having 4 modules of size 12.34 KLOC, 12.345 KLOC 12,345 LOC, 33.125 KLOC and 23.678 KLOC 23,678 LOC respectively. Consider the following cost drivers: High Skilled Programmer and Analyst, High Database Size and Very High Use of Software Tools. Hint : add LOCs of all the four modules after converting into KLOC format and the use that in the CoCoMo.

2. Suppose you are developing a software product of organic type. You have estimated the size of the product to be about 100 KLOC 100000 lines of code. Compute the nominal effort and the development time. Here, no cost drivers. Compute the nominal effort and development time alone.
No need to distribute the effort, duration and manpower as per this question.

3. A semidetached project size of 200 KLOC is to be developed. Software development team has average experience on similar type of projects. The project schedule is not very tight. Calculate the Effort, development time, average staff size.

4. A new project with estimated size of 40,000 LOC embedded system has to be developed. Project Manager of this project has a choice of hiring from two groups of developers A & B.

   a. Group A: Very high application experience, very little experience in the programming Language and very high skilled Analysts.
   b. Group B: Very low application experience, very high experience in the programming Language and very low skilled Analysts.

   Analyze and suggest to Project Manager with evidence, the impact of hiring the developers either from Group A or B. a=2.8, b=1.20, c=2.4, d=0.32.

5. Is it true that a software product can always be developed faster by having a larger development team? Assume that all developers are equally skilled & experienced.

6. Suppose a project was estimated to be 400 KLOC. Calculate the effort and development time for each of the three model i.e., organic, semi-detached & embedded. Use 400 KLOC and estimate the effort and the duration for all three types of project. Distributions & manpower calculation are not required

7. Suppose you are project manager of a large project. The top management informs that you would have to do with a fixed team size throughout the duration of project. What will be the impact of this decision on your project?